Synthetic Outpatient Clinic Database

Harmain Raza | ID: 25023333 | Github

How the data was generated

The data has been created using Python (sqlite3, random, and the Faker library) in a Google Colab notebook. No external datasets were used. Faker produced only fake names. All other values were selected from distributions that I selected. The database is completely reproducible due to the use of random seeds (random.seed(42), Faker. seed(42)).

Departments, Doctors, Medications, Patients, Appointments, and Prescriptions were filled in dependency order. The data are not independent, but correlated, in order to make the data realistic:

1. **Patients (1,500):** about 18% are children. In the case of children, their heights are normally distributed by age; otherwise, by gender. Weight is based on a random BMI and therefore is a tracking metric for height. The approximate real-world frequencies are used for blood types (O+ 35%, AB- 1%). Registration dates are never before birth.
2. **Appointments (5,000):** dates fall on weekdays, never before the patient's registration. Children attend Pediatrics and adults never do. Triage is determined by department. Wait time varies according to an exponential distribution: 46 min for Low triage, 5 min for Critical triage. Fees include department and doctor seniority/triage. As waiting increases, satisfaction decreases (4.15 for waits less than 15 min as opposed to 2.58 for waits of 45+ min). The sampling was unequal, with weights on patients to create frequent attenders (the busiest has 28 visits).
3. **Prescriptions (5,345):** 60.8% of visits produce one, with 1 to 3 drugs each, chosen by department (Psychiatry only receives antidepressants and anxiolytics). Child doses are proportional to body weight (child average dose is 281 mg, adult is 519 mg of Amoxicillin).
4. **Missing data:** To mimic real data, there are some missing data: a little over 24% of satisfaction ratings are NULL, 2% of temperatures, and 3% of adult smoking statuses.

Schema

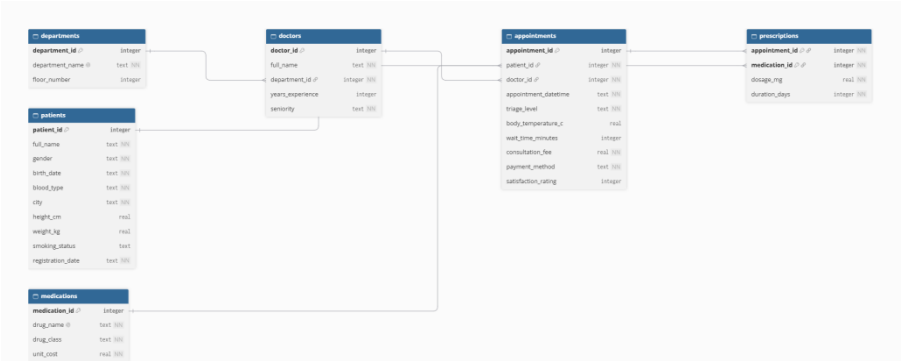


Figure 1: Entity relationship diagram

Table 1: Measurement scale of each Column

Scale	Columns	Reasoning
Nominal	gender, blood_type, city, department_name, drug_class, payment_method	Categories with no order
Ordinal	triage_level (Low < Medium < High < Critical), seniority, smoking_status, satisfaction_rating (1 to 5)	Ordered categories ;gaps between levels are not equal
Interval	body_temperature_c, appointment_datetime, birth_date, registration_date	Difference are meaningful but zero is arbitrary (0 °C, calendar zero)
Ratio	height_cm, weight_kg, consultation_fee, wait_time_minutes, years_experience, dosage_mg, duration_days, unit_cost	True zero (e.g a fee of 0 means a free visit), so ratios make sense

The database has 6 tables and 11,953 rows in total.

Table Design and Constraints

1. **Separate tables:** Department details are repeated only once per doctor row; for example, the department ID. This eliminates redundancy and update anomalies in patients, doctors, and medications. They are all one-to-many relationships (one patient may have many appointments).
2. **Compound key:** appointments and medications form a many-to-many relationship, resolved by the prescriptions junction table with PRIMARY KEY (appointment_id, medication_id). This prevents the same drug being prescribed twice in one appointment, and a test insert was rejected (UNIQUE constraint failed).
3. **Foreign keys:** doctors → departments, appointments → patients/doctors, prescriptions → appointments/medications. Enforced with PRAGMA foreign_keys = ON. PRAGMA foreign_key_check returned no violations, and a prescription for a non-existent appointment was rejected.
4. **CHECK constraints:** restrict the ordinal and nominal columns to valid categories, limit values to plausible ranges (e.g. weight 2 to 250 kg, rating 1 to 5), and require dosage_mg > 0 (a negative dose was rejected). **NOT NULL** is used on required fields, and **UNIQUE** on department and drug names.
5. **Indexes** on foreign key columns help to speed up joins. PRAGMA integrity_check returned ok.
6. **Zeros:** consultation_fee = 0 for free visits is a meaningful zero, which is why fee is ratio data, while 0 °C is arbitrary, so temperature is interval.

Ethics and Data Privacy

1. **No real individuals:** all names and records are fictional, and therefore not a threat to privacy. The names could be coincidental with real names, and should not be interpreted as having any meaning.
2. **If this were real:** health data is a “special category” of data within GDPR (UK/EU) and protected by HIPAA in the US, which must be justified by a lawful basis, explicit

consent, or a clinical purpose, and must be protected by strict access control, encryption, and security.

3. **Data minimization:** Design only stores city (not address), not phone/email, not national ID. The patients are identified by a pseudonymous integer ID.
4. **Re-identification risk:** The anonymity of the data would not prevent identification if they had also included birth date, city, and a rare diagnosis or medication (e.g., oncology drugs). The generalization (age bands), suppression of small groups, or k-anonymity checks would have to be applied to real data.
5. **Bias:** synthetic data is based on my ideas. However, distributions (e.g., only UK cities, 1% "Other" gender share, simplified blood-type frequencies) are unlikely to represent groups, and therefore models would not be able to generalize. It should not be used in actual clinical practice, as this represents an inappropriate use of the instrument.
6. **Transparency:** the code of the generator is open and has accompanying reasoning which is commented out, users can see exactly how the data was created.